

APEX Project Status (Implementation-Based)

This document describes **what the repository actually does today**, not a product roadmap. Details come from `backend/src` and `docs/vnv_alignment.md` (the latter is only a short strategy note; formal SRS/V&V PDFs in `docs/` are outside this audit).

Maturity

- **Backend:** FastAPI app with migrations, role-based admin/trainee routes, plugins, research export service.
- **Scoring:** Rule-based pipeline with optional LLM merge (`generate_feedback_hybrid`, `generate_feedback_hybrid_v2`) implemented in `ScoringService`; selectable via **Evaluator** plugin.
- **NLU:** `SimpleRuleNLU` + `NLUPipeline` / `turn_analysis` — heuristic, English keyword-driven behavior.

Stable Behaviors (Observable in Code)

- **Session lifecycle:** `SessionService.create_session` (reuse active session unless `force_new`); `close_session` sets `completed` + timestamps + `duration_seconds`.
- **Turn persistence:** User and assistant `Turn` rows with `spans_json`, `metrics_json`, `spikes_stage`.
- **SPIKES in dialogue:** `StageTracker` stages `setting ... strategy` persisted on `sessions.current_spikes_stage`.
- **Scoring endpoint:** `POST /v1/sessions/{session_id}:close` → `generate_feedback` → evaluator + metrics plugins.
- **Plugin registration:** Deterministic `PLUGIN_MODULES` import list + `PluginRegistry`.
- **Research:** Admin-only `/v1/research/*` with **deterministic anon session ids** (`ResearchService.generate_anon_session_id`).

Implemented Features (Non-Exhaustive)

- Text and **audio** turns (Whisper + `AudioToneAdapter`).
- Optional **assistant TTS** + storage (`dialogue_service._create_assistant_audio`).
- **WebSocket** dialogue: `/v1/ws/sessions/{session_id}` with JWT query param.
- **Trainee analytics:** `GET /v1/analytics/my-sessions` (`TraineeAnalyticsService`).
- **Admin:** session lists, case management, `/admin/plugin-registry`, `/admin/plugins` — see `admin_controller.py`.
- **Feedback entity** with rich JSON fields and `evaluator_meta` (includes `session_plugins` when set in `_persist_feedback_from_rule_state`).

Extensible (By Design)

Area	Mechanism
Patient replies	New <code>PatientModel</code> + <code>PLUGIN_MODULES</code> + settings/case

Area	Mechanism
Scoring methodology	New Evaluator delegating to new or existing ScoringService methods
Research metrics	New MetricsPlugin + session <code>metrics_plugins</code> list
LLM vendor (patient)	DefaultLLMPatientModel branches on default_llm_provider (<code>openai</code> vs <code>gemini</code>)

Known Limitations (From Current Implementation)

- **NLU / SPIKES detection** are keyword-heuristic (**StageTracker**, **SimpleRuleNLU**); no learned discourse parser in-repo.
- **Patient LLM path** in `sessions_controller` / `ws_controller` constructs `DialogueService(..., OpenAIAdapter(), ...)` while **DefaultLLMPatientModel** ignores that injected adapter and instantiates its own LLM from **settings** — can confuse readers of the HTTP layer.
- **Metrics plugins** only run in `generate_feedback`, not inside `generate_feedback_hybrid` / `_hybrid_v2` alone.
- **Session states** `paused` / `abandoned` exist on the entity but listing filters in `list_user_sessions` use `active` / `completed` only.
- **ApexMetrics** depends on **ScoringService private methods** — tight coupling for example metrics plugin.
- **Research anon id** reverse lookup scans batches of sessions (`resolve_anon_to_session_id`) — acceptable for small DBs; may need indexing strategy for very large deployments (not implemented).

“Research-Ready” (What Is Actually True)

- Structured **turn** + **feedback** storage suitable for CSV/JSON export.
- Admin-only export endpoints with **no raw session_id** in research JSON (anon ids).
- **Frozen plugin ids** on `sessions` for configuration traceability.
- **Evaluator meta** can carry compact LLM outputs and `session_plugins` for provenance on hybrid paths.

Out of Scope for This Spec

- Frontend feature list, hosting, IRB, or clinical validation claims.
- Future features **not** present in code (e.g. automatic scoring without `:close`, plugin filesystem auto-discovery, multimodal emotion models) — not documented here.